www.tntextbooks.in

d. Cell-A is sensitive to bright light intensities

5. Assertion: The imbalance in concentration of Na^+, K^+ and proteins generates action potential. Reason: To maintain the unequal distribution of Na^+ and K^+, the neurons use electrical energy.
   a. Both Assertion and Reason are true and Reason is the correct explanation of the Assertion.
   b. Both Assertion and Reason are true but the Reason is not the correct explanations of Assertion.
   c. Assertion is true, but Reason is false.
   d. Both Assertion and Reason are false.
6. Which part of the human brain is concerned with the regulation of body temperature?
   a. Cerebellum
   b. Cerebrum
   c. Medulla oblongata
   d. Hypothalamus
7. The respiratory centre is present in the
   a. Medulla oblongata
   b. Hypothalamus
   c. Cerebellum
   d. Thalamus
8. Match the following human spinal nerves in column I with their respective number in column II and choose the correct option

column I	column II
P. Cervical nerves	i. 5 pairs
Q. Thoracic nerve	ii. 1 pair
R. Lumbar nerve	iii. 12 pair
S. Coccygeal nerve	iv. 8 pair

   a. (P-iv),(Q-iii),(R-i),(S-ii)
   b. (P-iii), (Q-i), (R-ii), (S-iv)
   c. (P-iv),(Q-i),(R-ii),(S-iii)
   d. (P-ii), (Q-iv), (R-i), (S-iii)
9. Which of the following cranial nerve controls the movement of eye ball ?
   a. trochlear nerve b. optic nerve
   c. Olfactory nerve d. vagus nerve.
10. The abundant intracellular cation is
    a. H^+ b. K^+ c. Na^+ d. Ca^{++}
11. Which of the following statements is wrong regarding conduction of nerve impulse?
    a. In a resting neuron, the axonal membrane is more permeable to K^+ ions and nearly impermeable to Na^+ ions.
    b. Fluid outside the axon has a high concentration of Na^+ ions and low concentration of K^+, in a resting neuron.
    c. Ionic gradient s are maintained by Na^+ K^+ pumps across the resting membrane, which transport 3Na ions outwards for $2K^+$ into the cell.
    d. A neuron is polarized only when the outer surface of the axonal membrane possess a negative a charge and its inner surface is positively charged.
12. All of the following are associated with the myeline sheath except
    a. Faster conduction of nerve impulses
    b. Nodes of Ranvier forming gaps along the axon
    c. Increased energy output for nerve impulse conduction
    d. Saltatory conduction of action potential